Unit 5 – Polynomial Functions – Day 4 *Accelerated II/III*

Dividing Polynomials – Long Division

Review: $\frac{15}{2}$

New: $\frac{f(x)}{d(x)} = q(x) + \frac{r(x)}{d(x)}$

Example 1: Divide $2x^2 + 13x - 7$ by $x + 6$

Example 2: $(3x^2 + 11x + 1) \div (x - 3)$

Remainder Theorem: If a polynomial $f(x)$ is divided by $(x - k)$, then the remainder is _____

Look back at *Example 2*: $(3x^2 + 11x + 1) \div (x - 3)$

$f(x) =$ _____ , $(x - k) =$ _____ , $k =$ _____

Example 3: $(x^3 - 12x^2 - 42) \div (x^2 - 2x + 1)$